Development and Evaluation of a Macropore Flow Component for LEACHM

M. J. Borah, P. K. Kalita

Abstract. *LEACHM (Leaching Estimation And Chemistry Model) and a modified LEACHM were used to predict NO_3-N and atrazine transport in two Kansas soils. NO_3-N and atrazine concentration data obtained from two monolith weighing lysimeters containing two different Kansas soils were used to calibrate and validate the nitrogen and pesticide versions of LEACHM. Since LEACHM (ver. 3.0) does not include any means to deal with macropore flow, the macropore flow subroutine from RZWQM (Root Zone Water Quality Model) was incorporated into LEACHM. Measured concentrations of NO_3-N and atrazine were compared with those predicted by LEACHM with and without the macropore flow subroutine. In general, concentrations predicted by both models tend to follow the trend of measured concentrations. Addition of the macropore flow subroutine improved the predictions a little, especially for atrazine transport in silty clay loam soil. However, no such improvement was observed in sandy loam soil. Overall, the results of this study showed that addition of the macropore flow component had potential for improving prediction capability of LEACHM in silty clay loam soils. The use of more site specific macroporosity data is likely to improve the model predictions.*

Keywords. *Lysimeter, LEACHM, Modeling, NO_3-N, Atrazine, Transport.*

The ever growing demand for food and increase in the cost of living have forced today's farmers to continuously seek an improvement in crop production. This results in an increase in the use of chemicals in agricultural fields. Urea-Ammonium-Nitrate (UAN) fertilizer and atrazine are two widely used agrochemicals in Kansas. Nitrogen (N) is one of the most important nutrients required for plant growth and development. Losses of N to the subsurface environment can occur as a result of excessive fertilizer application, inefficient uptake of N by crops, and mineralization of soil N (Canter, 1996). Atrazine, on the other hand, helps to increase yield by improving nutrient availability to crops through better weed control. However, nitrate-nitrogen (NO_3-N) and atrazine leaching from agricultural fields are major sources of contamination for water resources in Kansas. A Kansas farmstead study (Steichen et al., 1988) indicated that of the 103 statistically representative samples collected statewide, 29 samples had nitrate levels above the EPA maximum contaminant level of 10 mg L^{-1} for drinking water and eight had detectable levels of pesticides. Atrazine was the only pesticide detected more than once.

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Computer simulation models have become popular tools for studying the transport mechanism of agricultural chemicals through subsurface zones. Simulation studies can be used as an inexpensive, time saving, and environmentally safe technique to evaluate the effects of various agricultural management practices on the subsurface movement of NO_3-N (Singh and Kanwar, 1995a). Over the past several years, a large number of computer simulation models have been developed. But few data sets are available for testing a range of models, few models have been tested on a range of soil types, and very few models have much demonstrable ability to simulate transient field leaching conditions (Addiscott and Wagenet, 1985). Few of the models developed so far are suitable for solving management related problems because of the lack of thorough validation (de Willigen et al., 1990). The models vary widely in their conceptual approach and degree of complexity, and are strongly influenced by the environment, training, and preoccupations of their developers (Addiscott and Wagenet, 1985). Models also vary in complexity, output presentation, and input parameter requirements. Spatial and temporal variations involved with the data result in a high degree of uncertainty associated with the results obtained (Verma et al., 1995). Before using a model for a particular soil type and environmental conditions, it is important to corroborate the model for these conditions.

The USDA-ARS (1996) developed RZWQM to simulate most of the important processes taking place in the root zone of an agricultural environment. The model also allows interaction of several important management practices and simulates their effects on the root zone. Singh and Kanwar (1995a) added a tile flow component to RZWQM for simulating subsurface drainage. They further enhanced RZWQM to simulate NO_3-N concentrations and NO_3-N losses with subsurface drainage (Singh and Kanwar, 1995b). Comparison with measured data showed that the modified model had a good potential for predicting NO_3-N concentrations and losses in the subsurface drain effluent under different tillage systems, but there was scope for further improvement of the model. Another comprehensive process based research model for simulating solute transport in the root zone is Leaching Estimation And Chemistry Model (LEACHM), developed by Hutson and Wagenet (1992). LEACHM consists of five submodels: LEACHW simulates soil-water flow in the subsurface zone; LEACHN simulates nitrate transport and transformations; LEACHP simulates pesticide transport and degradation; LEACHC describes transient movement of inorganic ions; LEACHB describes microbial population dynamics in the presence of a single growth supporting substrate. A study conducted by Khakural et al. (1995) showed that LEACHM made reasonable predictions of the depth of atrazine movement and atrazine concentrations in the soil profile for a sandy loam soil, but in silt loam and clay loam soils, depth of atrazine movement and atrazine concentrations in the soil profile were predicted reasonably well only during the relatively drier years. One of the major limitations of the use of LEACHM is lack of a macropore flow component. Previous evaluation of the Richard's and convection dispersion equations of LEACHM suggested that part of the simulation error was related to the inability of the model to simulate the macroporous water flow in well-structured soils (Jabro et al., 1995). Smith et al. (1991) attributed the inaccurate predictions of LEACHM and PRZM to simplistic linear adsorption equations and lack of a macropore flow subroutine. Inskeep et al. (1996) also reported that the lack of a preferential flow component is a possible reason for overestimation of mean travel time of pesticides by LEACHM. These studies suggest that incorporating a macropore flow component into LEACHM may enhance its performance.

The objectives of this study were to (1) incorporate the macropore flow subroutine and the water flow subroutine (during infiltration process) of RZWQM into LEACHM, and (2) use the modified LEACHM model to predict nitrate-N and atrazine transport in two Kansas soils in monolith weighing lysimeters. One of the soils used in this study was silty clay loam and vertical cracks were observed on the surface during dry periods. Therefore, macropore flow would likely play an important role in the transport of water and chemicals in this soil.

Materials and Methods

Field Experiments

These experiments were conducted at the North Agronomy Farm of Kansas State University, Manhattan, Kansas, since 1995. Details of the experiments can be found in Kalita et al. (1996, 1998). Four monolith (undisturbed) weighing lysimeters have been used for these field experiments. Two of the lysimeters contain Grundy silty clay loam soil from northeast Kansas and the other two lysimeters contain Clark sandy loam soil from south-central Kansas. The lysimeters were instrumented with solute suction samplers, tensiometers, soil temperature probes, neutron probe access tubes, loadcells, raingauges, and a datalogger. Figure 1a shows a schematic diagram of a lysimeter and the location of solute suction samplers, tensiometers, and neutron probe access tubes. During 1995-1997, the lysimeters were planted with corn (*Zea Mays* L.) at a conventional rate and the herbicides atrazine and alachlor were each applied at the rate of 2.0 kg/ha. Nitrogen fertilizer was applied at the rate of 168 kg N/ha in 1995 and 1996. But in 1997, it was applied at a rate of 659 kg N/ha due to a measurement error. Herbicides were surface broadcast on the day after planting while fertilizer was applied after emergence of the plants.

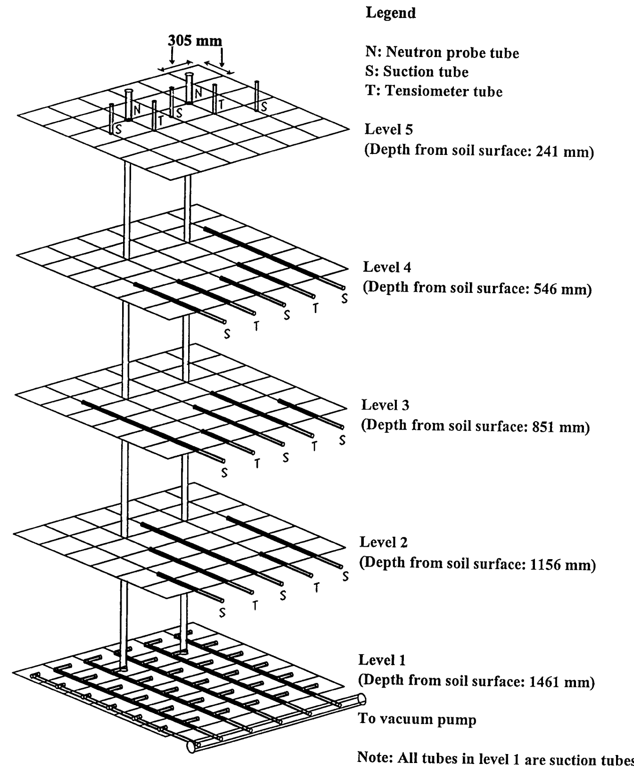


Figure l(a)–Isometric view of soil instrumentation.

Water samples were collected from each depth through the solute suction samplers. Twenty-four hours prior to sample collection, resident water (if any) was removed from the samplers and a vacuum pressure of 70 kPa was applied. In 1995, water samples were collected every week throughout the growing season (May through October). The water sample collection interval was increased to two weeks in the growing seasons of 1996 and 1997. Due to dry weather conditions, only a few samples were collected in 1996. Also, at depth 241 mm, very few samples were obtained in 1995 and no samples were obtained in 1996 and 1997. Water samples were analyzed for NO_3-N using Alpkem RFA methodology (ALPKEM Corporation, 1986). Atrazine concentrations were determined by using the Atrazine RaPID Assay (Ohmicron Environmental Diagnostics) method. We could not use gas chromatography (GC) because of insufficient volume of water samples.

Model Selection and Descriptions

Several models were reviewed before selecting LEACHM for simulating NO_3-N and atrazine concentrations in lysimeter conditions. Table 1 presents some of the models available for simulating agricultural chemical concentrations in soils and groundwater. We have

Table 1. Selected models for agricultural chemical transport

Model	Description	Performance
CREAMS Knisel (1980)	Simulates N, P, and pesticide transport in the root zone and predicts average concentrations of sediment-laden and dissolved chemicals in the runoff, sediment, and percolation.	Performance not satisfactory in simulating nitrogen leaching.
DRAINAGE Kanwar et al. (1983)	Simulates the soil-plant-water-nitrogen system in a typical tile-drained agricultural field.	Fairly capable of simulating NO_3-N concentrations in tile drain flow, but not suitable for simulating lysimeter conditions.
GLEAMS Leonard et al. (1987)	Evaluates the effects of agricultural management systems on the movement of agricultural chemicals within and through the plant root zone. A management tool to assess the impacts of BMPs and mainly concerned with surface runoff and erosion processes.	Performed reasonably well in predicting pesticide movement and leaching.
GLEAMS-DRAINMOD Thooko et al. (1994)	Pesticide component of GLEAMS has been integrated to DRAINMOD to simulate subsurface drainage water quality especially pesticide concentrations. It was primarily intended for simulating subsurface drainage water quality.	Initial evaluation was not satisfactory, and also not suitable for simulating lysimeter conditions.
LEACHM Hutson and Wagenet (1992)	Simulates nitrogen transport and transformation, pesticide degradation and displacement, transient movement of inorganic ions, and microbial population dynamics.	Performance is reasonably good as demonstrated by several researchers.
NCSWAP Molina and Richards (1984)	Simulates N transformation and transport processes in the subsurface zone, plant uptake of N and water, crop yield, root growth, and N fertilizer dynamics.	Performance was found to be satisfactory. Statistical performance of LEACHM was found to be better than NCSWAP.
PESTAN Enfield et al. (1982)	Estimates concentrations of pesticides in the unsaturated zone. It uses an average recharge rate rather than considering the effect of rainfall, irrigation, and ET independently.	Makes reasonable predictions of leaching, but not suitable for simulating lysimeter conditions.
PESTFADE Clemente et al. (1993)	Simulates simultaneous movements of water and solutes in the soil environment as affected by runoff, leaching, dispersion, uptake, macropore flow, sorption/desorption, volatilization, heat flow, and chemical and microbial degradation.	Performed well in predicting atrazine concentrations in 0-20 cm depth as shown by a field study conducted by the authors in Canada. In the same study, it was found to perform better than GLEAMS.
PRZM Carsel et al. (1984)	Simulates pesticide transport in surface runoff and in the vadoze zone including plant uptake, sorption, and biochemical degradation of pesticides.	Satisfactory overall performance as demonstrated by a large number of researchers. LEACHM was found to perform better than PRZM.
RZWQM USDA-ARS (1996)	Simulates plant growth and the movement of water, nutrients and agro-chemicals over, within, and below the crop root zone of a unit area of an agricultural cropping system under a range of common management practices.	Comprehensive, well-documented, and validated by a number of researchers under different conditions, but highly data intensive and very complex to use.
SESOIL Bonazountas and Wagner (1984)	Predicts solute distribution in both the soil profile and watershed on a seasonal basis	Performance not satisfactory.
TDNIT Bogardi and Bardossy (1984)	Simulates the movement of water and nitrogen through the unsaturated zone and also estimates the amount transported to groundwater.	Low input data requirement and suitable for use in Monte-Carlo simulation procedures. Very few evaluations are available.

studied some of these models based on their suitability for simulating lysimeter conditions. During the model selection process, the following three factors were considered: ability to simulate a closed bottom boundary condition; availability of model components for simulating major physical, chemical, and biological processes in the root zone; and ease of use. An attempt was also made to avoid models that simulate both surface and subsurface flow as this would involve additional computations and complexities (simulating the processes in the root zone is the primary interest of our lysimeter study). Based on these factors, LEACHM was selected for this study.

LEACHM uses a finite difference form of the one dimensional Richard's transient flow equation for predicting water fluxes and soil moisture distribution with time. Water retention and conductivity functions are calculated by the methods proposed by Campbell (1974). A one dimensional convection-diffusion equation (CDE) is solved numerically to estimate chemical fluxes and distribution in the profile. Crop growth subroutines used in

LEACHM are based upon empirical equations and the effects of water content, soil strength, and nutrient concentration on root and shoot development are not considered. However, the distribution of roots with depth partly determines the water and chemical uptake terms within each soil segment. Details of the model can be found in Hutson and Wagenet (1992).

RZWQM is a one-dimensional, deterministic, process-based field-scale research model. The model consists of subroutines for simulating water, chemical, and heat transport, plant growth, evapotranspiration, soil chemistry in the root zone, pesticide transport and degradation, organic matter (OM), and nitrogen cycling. RZWQM divides the water flow process into two phases (USDA-ARS, 1992). The first phase consists of infiltration into the soil matrix and macropores and macropore-matrix interaction during a rainfall or an irrigation modeled by using the Green-Ampt approach. The second phase consists of redistribution of water in the soil matrix, following infiltration. For modeling the redistribution process, RZWQM uses a mixed form of Richard's equation, unlike LEACHM which uses an h-based form of Richard's equation. These two forms of Richard's equation are described in detail in Celia et al. (1990). Also, the chemical transport in RZWQM is done by using partial piston displacement followed by mixing in small depth increments. More information about RZWQM can be obtained from the user's manual and technical documentation of RZWQM published by USDA-ARS (1992).

Incorporation of Macropore Flow Component into LEACHM

In this study, the macropore flow component from RZWQM was added to LEACHM and the modified LEACHM is described as MLEACHM. The incorporation process is described in the following sections.

In LEACHM (ver. 3.0), the entire soil profile is divided into a number of horizontal segments of equal thickness and several different criteria are used to divide the total time into short time intervals which are not necessarily of equal length. Subroutine WATFLO in the LEACHW submodel of LEACHM simulates flow and redistribution of water in the soil by numerically solving Richard's equation during a time increment. The Crank-Nicholson (1947) implicit method is used to solve the finite difference form of Richard's equation. A flux controlled upper surface boundary exists all the time, and the flux changes both in magnitude and direction based on ponded or nonponded infiltration, evaporation, or no change in moisture content. The lower boundary condition can be changed to represent a fixed depth water table, free draining profile or zero flux. The standard version of LEACHM assumes a horizontally homogenous soil profile which does not have strong structure, preferential flow paths such as cracks or worm holes, or display fingering phenomena.

The present modification of LEACHM involves transferring control from the subroutine WATFLO to the subroutine EVNTRO of RZWQM during a ponded or non ponded infiltration event caused by rainfall or irrigation (fig. 1b). As soon as the program enters the subroutine WATFLO, it checks for the upper boundary condition. If it is that of evaporation or zero flux, then it will carry out the remaining instructions (soil moisture redistribution based on Richard's equation) of the subroutine WATFLO.

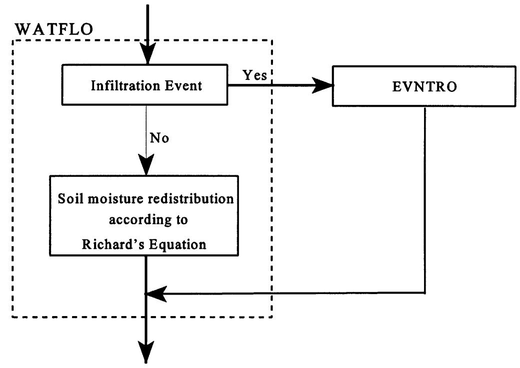


Figure 1(b)–Graphical representation of interaction between WATFLO and EVNTRO.

Therefore, there is no difference between operations of LEACHM and MLEACHM during evaporation or zero flux upper boundary condition. But when the upper boundary condition is that of an infiltration event, the program leaves WATFLO without carrying out the remaining instructions and moves on to the subroutine EVNTRO via the subroutine INTRFAC. EVNTRO is the main driver for infiltration/storm event hydrology for RZWQM. EVNTRO takes care of both water and chemical transport through the soil matrix as well as through the macropores (in case of an overland flow). After the infiltration event is over, the program again comes back to WATFLO and immediately moves on to the next subroutine without performing any operation in WATFLO. Also, at the end of the infiltration event, the overall time is advanced by an amount equal to the duration of the event. No other operations such as evapotranspiration, plant growth, and heat flow are performed by LEACHM during the infiltration event.

During an infiltration event , EVNTRO simulates water and chemical transport using layers of 10-mm thickness. EVNTRO has a macropore flow subroutine for simulating macropore flow during an infiltration event. Macropore flow takes place during an overland flow, which is generated when rainfall rate exceeds infiltration rate. Overland flow picks up chemicals from the surface soil through mixing and raindrop impact and makes them available for macropore flow. The water solution moving through macropores is subjected to lateral absorption into the drier soil matrix. The reactive chemicals in the solution are also subjected to adsorption to or desorption from the macropore walls. The top layers are assumed to have circular macropores which may change into planar cracks in the lower layers. A certain fraction of macropores in each layer are considered as deadend and assumed to branch horizontally off the continuous macropores in each layer. Average volume fraction of macroporosity and average size of voids are assumed to be known. This knowledge is used to calculate the number of pores and or total length of cracks per unit area of soil. During an overland flow, the water and solutes available at the soil surface are allowed to flow into macropores to the limit of macropore flow capacity. For each time step, the flow is sequentially routed downward through the continuos macropores in 1-cm depth increments. In each depth increment, the macropore flow is allowed to flow into the deadend macropores, and be absorbed by the soil matrix by radial or lateral infiltration from both continuous and deadend parts of macropores. The deadend macropores can also store the solution. The time dependent lateral absorption of the solution occurs only in the drier soil matrix below the transient wetting front generated by the continuing vertical infiltration into the matrix. Before the water and solutes flow into the deadend macropores or are absorbed into the soil matrix, the reactive chemicals in the available macropore flow equilibrate with the wall of the pore. This mixing and equilibration of macropore solution with the wall occurs in the wetted zone above the wetting front even though there is no lateral absorption into the soil matrix in this zone (USDA-ARS, 1992).

In RZWQM, soil properties and hydraulic properties are provided as input data for different horizons which are assigned to numerical layers (used during redistribution) and infiltration layers (used during infiltration) accordingly by the model. EVNTRO is concerned only with the infiltration layers as it is a driver for infiltration events only. Layer thickness for LEACHM is chosen as an integral multiple of infiltration layer thickness of RZWQM so that the soil and hydraulic property values from LEACHM can be easily assigned to the corresponding infiltration layers of RZWQM. This is done in a newly written subroutine READ2N which reads the additional input parameters for EVNTRO. Most of the input parameters used in EVNTRO are obtained directly from LEACHM. But some parameters such as those related to macroporosity and bottom boundary condition are provided through a separate input file. Another newly written subroutine, INTRFAC links WATFLO and EVNTRO by interchanging water content and chemical concentration between the grids of LEACHM and EVNTRO. It also links other relevant parameters between WATFLO and EVNTRO. RZWQM does not have an option for closed bottom boundary, but it does have an option for a constant flux bottom boundary. If a zero flux bottom boundary condition exists in LEACHM, the bottom boundary condition in EVNTRO is set to a constant flux boundary with an extremely small leakage rate. LEACHM and EVNTRO use different methods for calculating water retention and conductivity parameters. While LEACHM uses a modified Campbell method (Hutson and Cass, 1987), EVNTRO uses retentivity parameters based on functional forms suggested by Brooks and Corey (1964) with slight modifications (USDA-ARS, 1992).

Model Input Data

Input data required for the modified LEACHM (MLEACHM) can be divided into several categories:

Climatic Data. LEACHM requires daily rainfall data along with the starting time and rainfall rate and MLEACHM uses breakpoint rainfall data. Although, LEACHM requires daily rainfall data, it also supports the use of more than one rainfall event in a day provided there is no conflict in the duration of the events. Rainfall data used in this study consist of starting time and rainfall amount and rate. Rainfall rate for an event is determined from the amount and duration of the rainfall (obtained from tipping bucket raingauge). This is based on the assumption that the rainfall intensity remained constant for the entire duration. Irrigation events were also included in the rainfall data. Although, the irrigation rates were estimated based on the time and amount of water application, these values were also subjected to minor calibration. Since irrigation was applied manually, rainfall and irrigation never took place at the same time. The same rainfall data were used for both the models. But in MLEACHM, INTRFAC changes the format of rainfall data into break point rainfall data (consisting of two point, the beginning and end) before transferring the control to EVNTRO. Evapotranspiration (ET) data were based on changes in the weight of the lysimeter boxes obtained from hourly load cell readings. Since ET was estimated hourly, the effect of weight change due to crop growth on the lysimeter on an hourly basis was neglected. The data on change in the weight of the lysimeter were not used anywhere else as model input. Initial temperature and mean weekly temperatures and amplitudes were obtained from temperature probes installed at various depths in the lysimeters. In 1995 and during the first half of 1996 (when the datalogging system was not fully operational), climatic data were obtained from a weather station located within 200 m of the experimental site.

Soil Physical and Hydraulic Properties. Soil textural properties and soil organic matter at different layers were determined in the laboratory using the soil cores removed from the lysimeter soil profile during the installation of the solute suction samplers. Bulk density values for different layers of soils were determined from undisturbed cores obtained from the original sites. Both LEACHM and RZWQM estimate porosity based on bulk density and soil texture data. But, it was found that LEACHM estimated a porosity of less than 0.44 for the silty clay loam soil, while the measured soil moisture retention curve indicated a porosity of more than 0.50. Therefore, it was decided to provide porosity as direct input for these soils. Saturated hydraulic conductivity for the silty clay loam soil was obtained from previous works on these soils; whereas, that for the sandy loam soil was determined from undisturbed cores by constant head method in the laboratory. Soil moisture retention curves for different layers of both the soils were determined in the laboratory by using the pressure plate apparatus. Several points from such a curve were used as input data in subroutine RETFIT of LEACHM to estimate the Campbell retention parameters for the corresponding soil layers. Soil moisture retention parameters used in EVNTRO were obtained from the user's manual of RZWQM. During the winter seasons of 1995-1997, the lysimeters were almost fully saturated with water in December and January (mostly caused by winter rains, snow melt, and extremely low evaporation). Therefore, the initial soil-water distributions for both the soils were based on the assumption that the soil profiles were fully saturated on 1 January 1995. Soil-water in the lysimeters was measured using a neutron moisture depth probe and meter (503 DR Hydroprobe, Campbell Pacific Nuclear Corporation, Pacheco, Calif.). Although, tensiometer readings were also taken, only neutron probe data had been analyzed and used in this study.

Model Evaluation

The model evaluation consists of calibration and validation of the model. The calibration process involves the adjustment of selected model parameters within an expected range so as to minimize the discrepancies between observed and predicted values. Validation is essentially an independent test of the model, where the model predictions are compared with data not used in the calibration process (Donnigan, 1983). Evaluation of model performance should include both statistical criteria and graphical displays. A model is a good representation of reality only if it can be used to predict, within a calibrated and validated range, an observable phenomenon with acceptable accuracy and precision (Loague and Green, 1991).

Model calibration was carried out in two steps. During the first step, soil-water flow parameters were calibrated. During the second step (which involved calibration of parameters used in transport and transformation/ degradation of nitrogen and atrazine), parameters calibrated in the first step were kept unchanged. For simulation purposes, the lysimeter soil profile was divided into 10 different horizons, each of which might have different physical, hydraulic and chemical properties. Soil-water flow and chemical transport parameters were calibrated for each horizon. The calibration process was carried out starting from the top horizon and then proceeding toward the bottom horizon.

Volumetric water contents at eight different depths of the soil profile were available during growing seasons of 1996 and 1997 for the silty clay loam soil and during the growing season of 1997 for the sandy loam soil. Water content data were obtained at an interval of 7 to 10 days by using a neutron probe. Soil-water flow parameters (saturated hydraulic conductivity, porosity, soil moisture retention characteristics, total macroporosity, pan factor) were calibrated using all but two days of the 1997 volumetric water content data. Predicted water content values were verified using the water content data of those two days. For the silty clay loam soil, 1996 water content data were also used for verification of the model predictions. The same initial soil-water profile was used for both the models. In general, both LEACHM and MLEACHM predicted water contents well in both the soils. In the bottom layers of the sandy loam soil, MLEACHM underpredicted water content whereas LEACHM overpredicted it. A typical example is shown in figure 2a. Figure 2b shows measured and predicted water content in 1997 for the sandy loam soil. Both the models underpredicted water content in the bottom layers of silty clay loam soil, however, MLEACHM predictions were relatively closer to the measured values as shown in the example in figure 3a. Figure 3b shows measured and predicted water content in 1997 for the silty clay loam soil. The results of linear regression indicated that MLEACHM predictions were better than LEACHM in the silty clay loam soil (fig. 3b) whereas LEACHM predictions were relatively better in the sandy loam soil (fig. 2b).

Atrazine transport and degradation parameters were calibrated using the 1995 data set for the silty clay loam soil and the 1997 data set for the sandy loam soil. The remaining two years of measured data were used for validating the model predictions. Based on the past management history of the lysimeter soils, initial atrazine concentrations for both soils and both models were considered to be zero.

The NO_3-N transport and transformation parameters were calibrated using the 1995 data set for both soils. Validation was done by using the remaining two years (1996-97) of measured data. For NO_3-N transport parameters, calibration was started using the values given in the sample files of the LEACHM model (because those values were found to be within the stated range in the

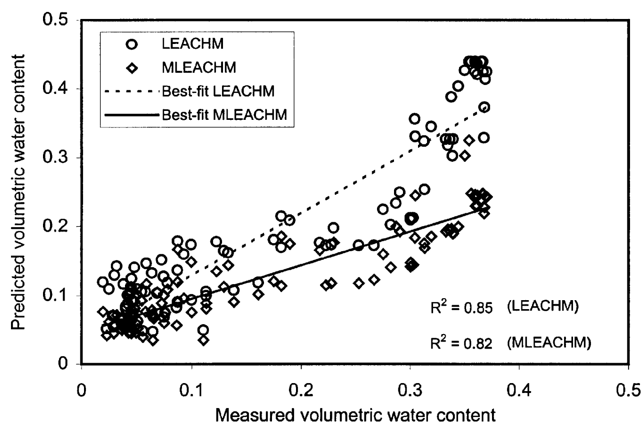


Figure 2(b)–Measured and predicted soil-water content in sandy loam soil in 1997.

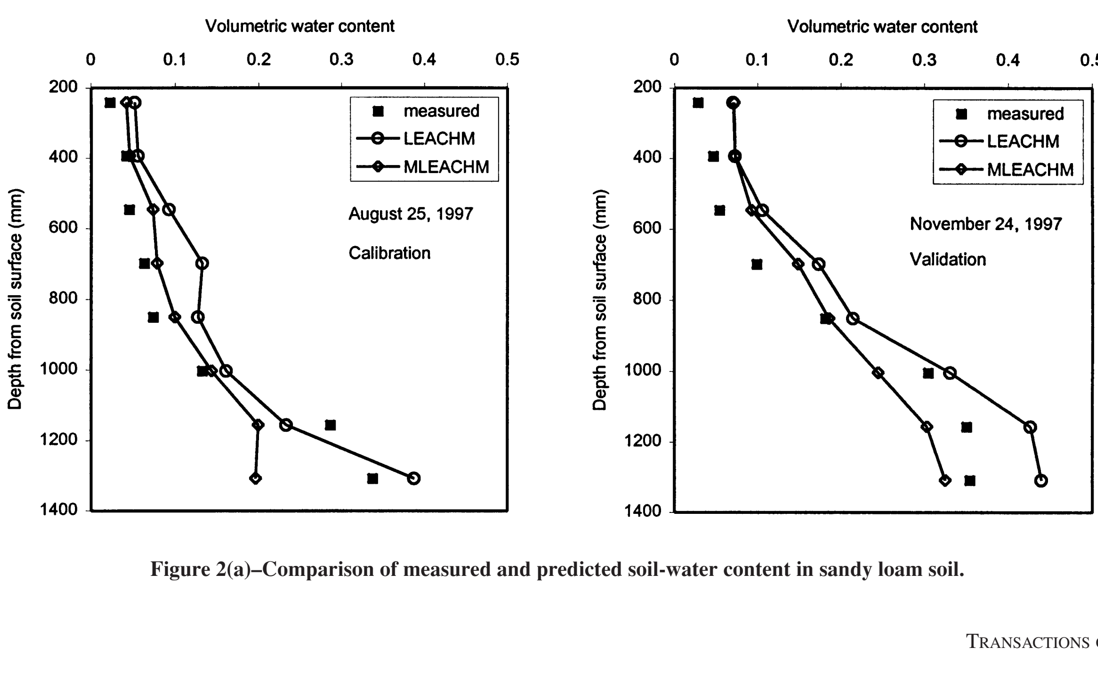


Figure 2(a)–Comparison of measured and predicted soil-water content in sandy loam soil.

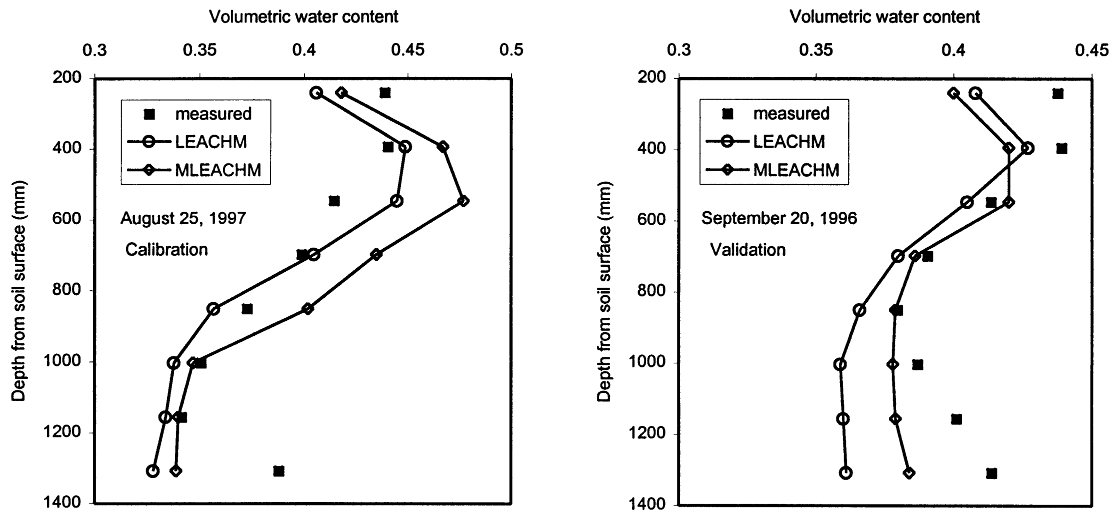


Figure 3(a)–Comparison of measured and predicted soil-water content in silty clay loam soil.

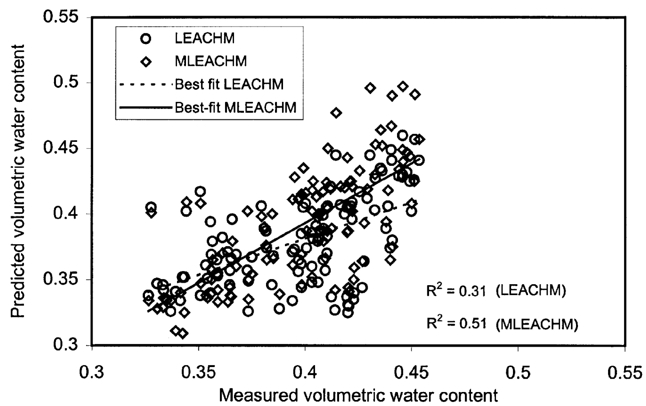


Figure 3(b)–Measured and predicted soil-water content in silty clay loam soil in 1997.

literature). Minor adjustments were made in the K_d value for NO_3-N, urea hydrolysis, nitrification, and denitrification rates. All other NO_3-N transport parameters were kept unchanged. The reason for calibrating K_d for NO_3-N (which is mostly considered as zero due to high mobility of NO_3-N) is that high concentrations of NO_3-N were observed in soil cores removed from the lysimeters indicating adsorption of NO_3-N in the soil. Initial nitrogen and carbon pools in the soil profiles in 1995 were estimated through calibration by using LEACHM. The same initial values were also used for MLEACHM simulations because a calibration using MLEACHM showed only minor changes in these values.

Table 2 gives the values of calibrated soil-water flow and NO_3-N and atrazine transport parameters for these two soil types in Kansas. Some of these parameters had different values at different depths for the same soil profile and the range of values are shown in the table.

Simulation accuracy of the model was evaluated based on its ability to predict nitrate-N and atrazine concentrations in soil-water at different depths from the soil surface. In addition to graphical displays of measured and simulated concentrations, the statistical methodologies suggested by Loague and Green (1991) were used to evaluate the prediction capabilities of LEACHM and MLEACHM. These statistical measures included root mean square error (RMSE) and modeling efficiency (EF) as follows:

Table 2. Selected input parameters after calibration

Parameter	Silty Clay Loam	Sandy Loam
Saturated hydraulic conductivity (mm day^{-1})	5-15	100-5100
Total macroporosity (10^{-5} cm^3 cm^{-3})	0.225-45	0.01-50
SFCT*	0.98	1.0
K_d for NO_3-N (L kg^{-1})	0.0	0.05
Urea hydrolysis rate (day^{-1})	0.16-0.36	0.16-0.36
Nitrification rate (day^{-1})	0.20	0.20
Denitrification rate (day^{-1})	0.06	0.10†
Litter mineralization rate (day^{-1})	0.01	0.01
Manure mineralization rate (day^{-1})	0.02	0.02
Humus mineralization rate (10^{-4} day^{-1})	0.70	0.70
Atrazine degradation rate (day^{-1})	0.0001-0.005‡	0.0001-0.024§
K_{oc}‖ (L kg^{-1}) for atrazine	163	163

* A factor which controls the amount of water and chemicals that can be absorbed from the macropores into the soil matrix during macropore flow.
† A rate of 0.05-0.30 was used for MLEACHM.
‡ A rate of 0.001-0.0155 was used for MLEACHM.
§ A rate of 0.08-0.14 was used for MLEACHM.
‖ Organic carbon partition coefficient.

$$RMSE = \left[\frac{\sum (P_i - O_i)^2}{n}\right]^{\frac{1}{2}} x\left(\frac{100}{\overline{O}}\right) \quad (1)$$

$$EF = 1 - \frac{\sum (P_i - O_i)^2}{\sum (O_i - \overline{O})^2} \quad (2)$$

where

P_i = simulated value
O_i = observed value
$\overline{O}$ = observed mean
n = number of observations

RMSE is a measure of the deviation of the simulated from the measured value and ideally it should be 0. An ideal value for EF is 1. If EF is less than zero, the model predicted values are worse than simply using the observed mean (Loague and Green, 1991). RMSE or EF values alone are not enough to evaluate or compare model performances. A lower RMSE value and a higher EF value indicate a better agreement between measured and simulated values, but it is necessary to verify this with graphical results. It is also not possible to define particular limits of these values for the acceptability of the results of the models. It is up to the user to decide what values are acceptable for the specific job for which the model is used. Based on the definition of RMSE and EF, if a model shows a better RMSE value, then it will also have a better EF value. Therefore, if LEACHM has a better RMSE value for a particular case, it will also have a better EF value for that case.

Results and Discussion

Results of the NO_3-N and atrazine simulations are presented using original LEACHM (without a macropore flow component) and with the modified LEACHM (incorporating the macropore component of RZWQM into LEACHM). For both NO_3-N and atrazine, calibration and validation were performed for all the four sampling depths, but, graphical results were presented only for selected depths and years (figs. 4 to 11). However, statistical comparisons of measured and simulated NO_3-N and atrazine concentrations were presented for all the depths and years (tables 3 and 4). Although experiments were conducted in four lysimeters, the measured data from only two lysimeters (one for each type of soil) were used for model evaluation.

Statistical analyses showed that prediction of NO_3-N concentrations by MLEACHM was relatively better than LEACHM in 8 out of 12 cases in the sandy loam soil and in 5 out of 11 cases in the silty clay loam soil. In sandy loam soil, MLEACHM performed better than LEACHM in 1995 and 1996, but in silty clay loam soil, performance of MLEACHM was better only in 1997. LEACHM performed better than MLEACHM in predicting atrazine concentrations in 8 out of 10 cases in sandy loam soil and in 2 out of 12 cases in silty clay loam soil. Both MLEACHM and LEACHM predicted zero concentrations at depths 1156 and 1461 mm in sandy loam soil in 1995. Therefore, the RMSE and EF values were equal for both the models in these two cases.

Table 3. Statistical comparisons of predicted and measured NO_3-N concentrations

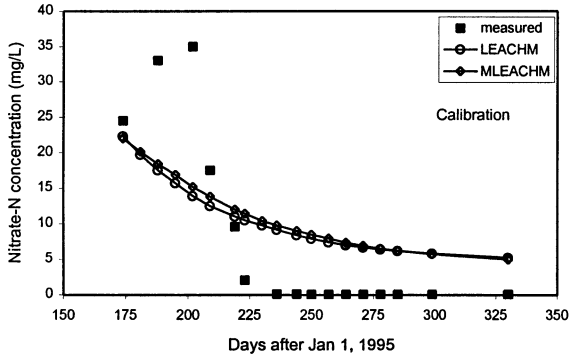
Para-meters	Depth (mm)	1995 LEACHM	1995 ML'M*	1996 LEACHM	1996 ML'M	1997 LEACHM	1997 ML'M
Sandy Loam							
RMSE	546	310.63	247.49	348.23	243.17	258.62	269.32
	851	150.39	121.92	86.24	87.67	118.80	122.62
	1156	117.43	116.38	95.95	92.46	76.35	77.88
	1461	139.67	73.08	187.16	44.97	138.43	115.90
EF	546	–1.58	–0.64	–5.98	–2.40	–2.12	–2.39
	851	0.28	0.53	–1.40	–1.48	–0.21	–0.28
	1156	0.46	0.47	–0.58	–0.47	0.59	0.58
	1461	–1.79	0.23	–27.48	–0.64	–0.43	–0.003
Silty Clay Loam							
RMSE	546	37.79	44.15	-	-	78.89	67.39
	851	107.06	110.71	114.59	114.85	120.58	83.53
	1156	98.91	98.88	195.66	196.76	176.56	174.14
	1461	94.30	94.14	73.56	83.20	116.84	117.81
EF	546	0.90	0.87	-	-	0.48	0.62
	851	0.08	0.02	0.01	0.006	–0.67	0.19
	1156	–0.27	–0.27	–0.16	–0.18	–0.38	–0.34
	1461	–0.09	–0.08	–0.29	–0.65	–2.21	–2.26

* ML'M = MLEACHM.

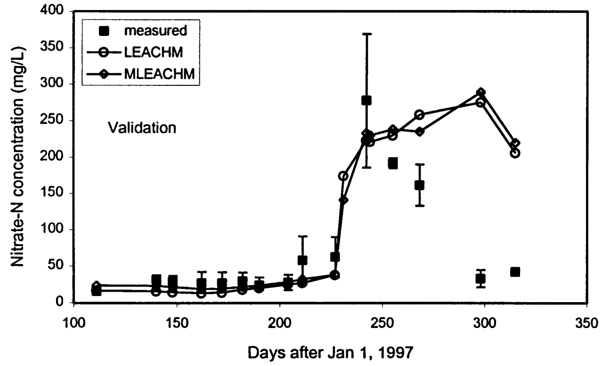
Table 4. Statistical comparisons of predicted and measured atrazine concentrations

Para-meters	Depth (mm)	1995 LEACHM	1995 ML'M*	1996 LEACHM	1996 ML'M	1997 LEACHM	1997 ML'M
Sandy Loam							
RMSE	546	215.25	180.40	362.10	531.62	52.12	124.01
	851	147.68	150.64	53.34	90.09	149.10	465.96
	1156	177.29	177.29	90.97	110.70	190.36	221.09
	1461	182.12	182.12	105.34	107.07	73.65	45.32
EF	546	–6.06	–3.96	–19.51	–43.11	0.82	–0.004
	851	–0.66	–0.73	0.19	–1.30	0.03	–8.42
	1156	–0.46	–0.46	–1.20	–2.25	–0.09	–0.47
	1461	–0.43	–0.43	–4.75	–4.94	–1.40	0.08
Silty Clay Loam							
RMSE	546	122.40	41.83	84.89	183.89	113.42	126.92
	851	130.60	113.93	146.36	129.03	98.94	42.85
	1156	131.98	121.60	250.05	234.02	123.13	72.51
	1461	138.47	137.10	128.06	98.65	161.08	154.89
EF	546	–1.99	0.64	–0.31	–5.17	–0.64	–1.06
	851	–1.42	–0.84	–0.87	–0.45	–8.61	–0.80
	1156	–1.34	–0.99	–0.19	–0.04	–1.93	–0.01
	1461	–1.09	–1.04	–1.56	–0.52	–0.62	–0.50

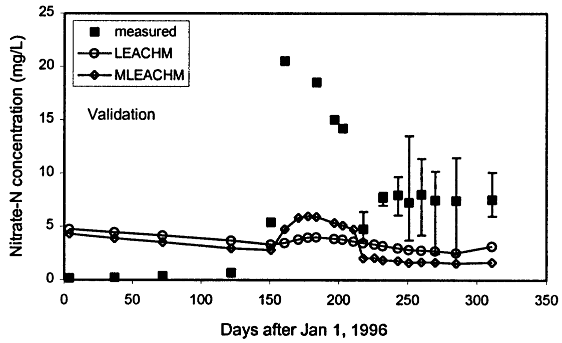
The NO_3-N concentrations predicted by both the models followed the trend of the measured NO_3-N concentrations for both the sandy loam and the silty clay loam soils fairly well (figs. 4 and 5). Both the models overpredicted at depths 546 and 851 mm in the sandy loam soil toward the end of the growing seasons during all three years. But, at depths 1156 and 1461 mm, both the models underpredicted NO_3-N concentrations in 1996-1997. In the silty clay loam soil, both the models seem to underpredict these concentrations most of the time. The performance of both the models were similar in the sandy loam soil except at depth 1461 mm where, MLEACHM performed slightly better than LEACHM. Similar results were indicated by the statistical analyses (table 3). Both the models performed similarly in the silty clay loam soil. No overland flow was observed in the sandy loam soil throughout the simulation process. The difference in the infiltration process between the simulations of the two models is that the original model used Richard's equations for water flow and the convection dispersion equation for solute transport; whereas, the modified model used Green-Ampt's equation and the partial piston displacement technique, respectively. The high concentrations at the beginning of 1996 as shown in figures 4b and 5c were due to overpredicted concentrations at the end of 1995 which were used as initial concentrations for 1996. The sudden increase in the measured NO_3-N concentrations in the growing season of 1997 (as shown in figures 4c, 4d, 5b, and 5d) was caused by a higher fertilizer application rate in that year. In the sandy loam soil, both the models were able to predict this rapid increase in concentrations at depths 546, 851, and 1156 mm, but they predicted a more gradual increase at depth 1461 mm. In the silty clay loam soil, both the models predicted a sudden rise in concentrations at depth 546 mm.



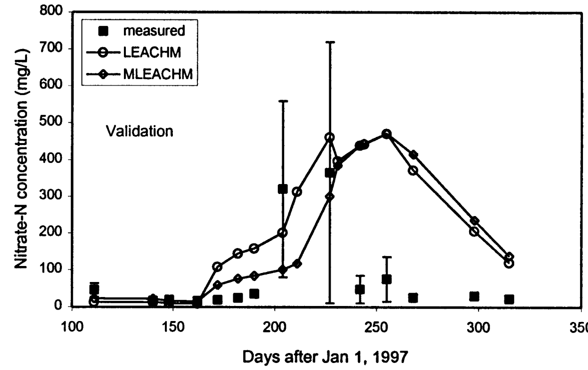
(a) Depth from soil surface = 1156 mm



(c) Depth = 851 mm

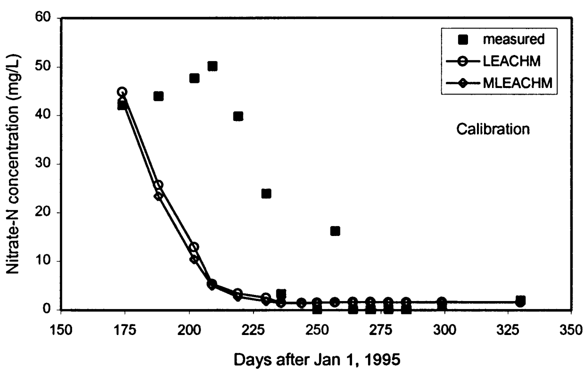


(b) Depth = 1156 mm

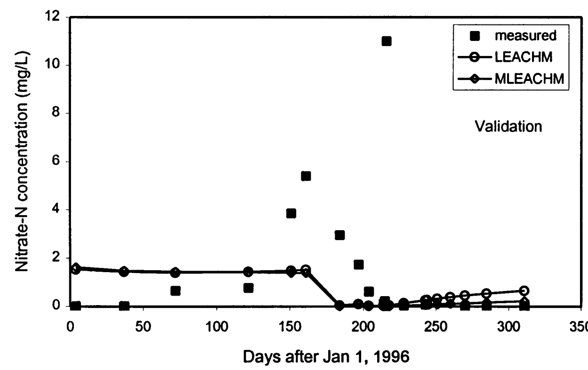


(d) Depth = 546 mm

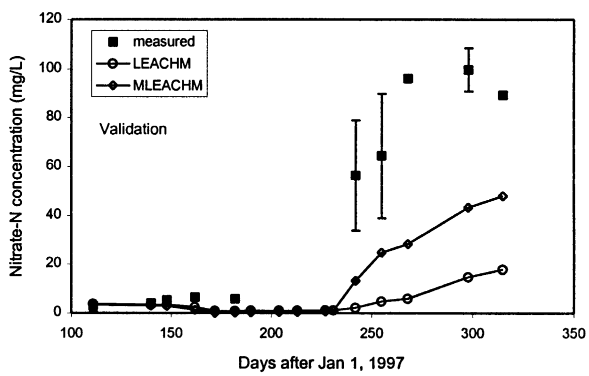
Figure 4–Comparison of measured and predicted nitrate-N concentrations in sandy loam soil.



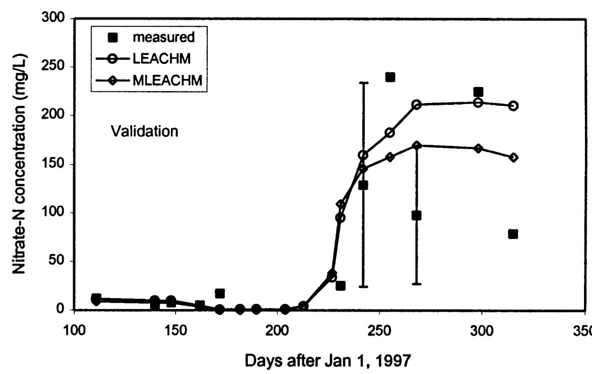
(a) Depth from soil surface = 851 mm



(c) Depth = 1156 mm

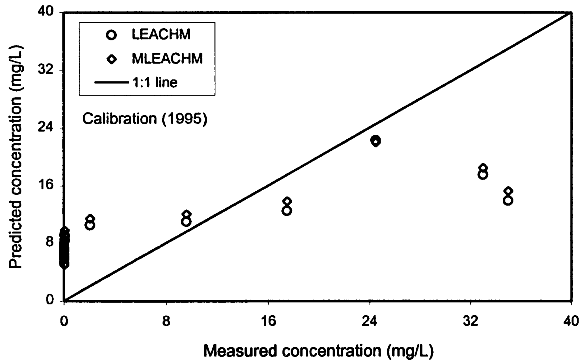


(b) Depth = 851 mm

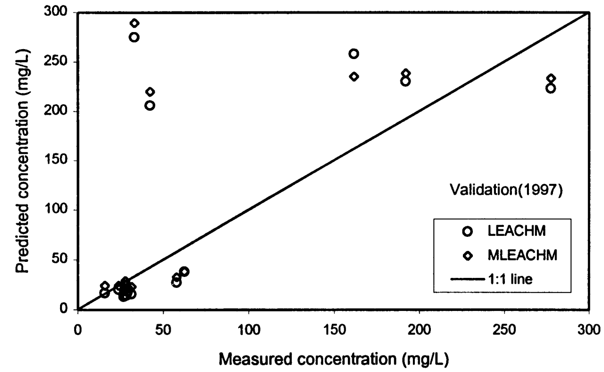


(d) Depth = 546 mm

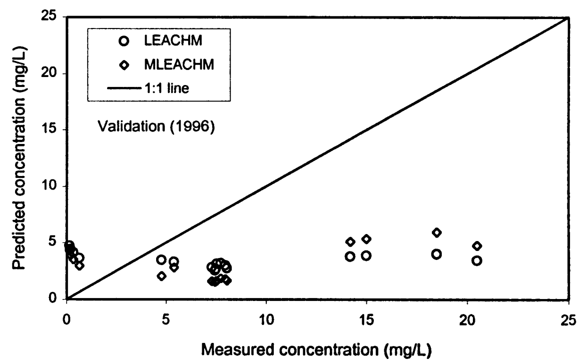
Figure 5–Comparison of measured and predicted nitrate-N concentrations in silty clay loam soil.



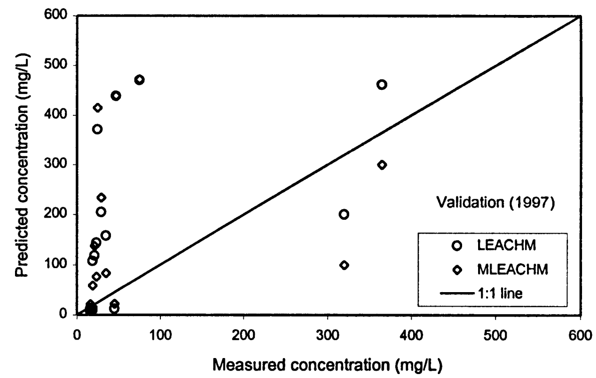
(a) Depth from soil surface = 1156 mm



(c) Depth = 851 mm

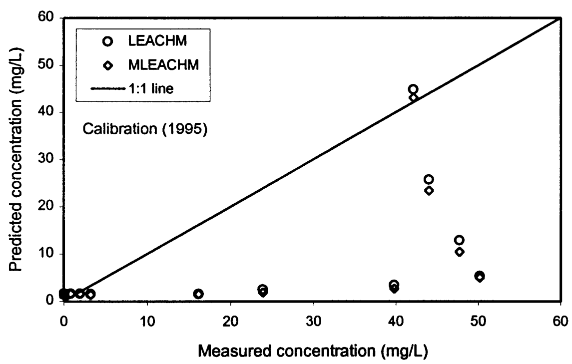


(b) Depth = 1156 mm

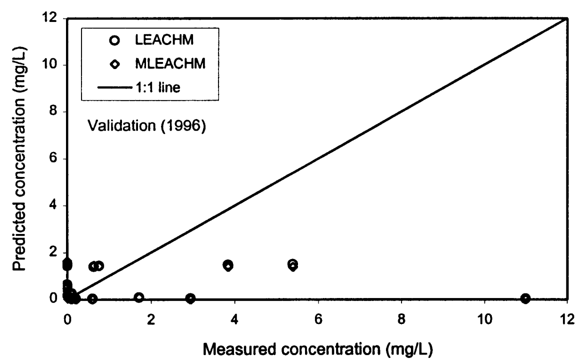


(d) Depth = 546 mm

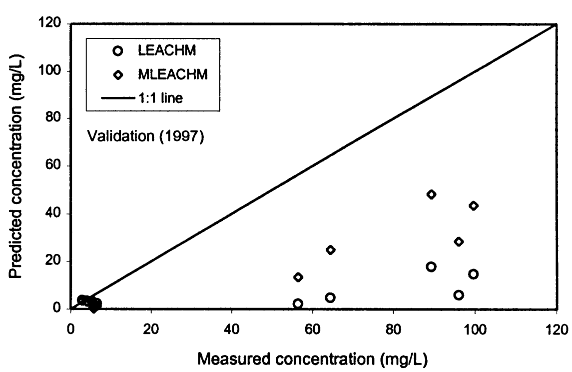
Figure 6–Comparison between measured and predicted nitrate-N concentrations and 1:1 line for sandy loam soil.



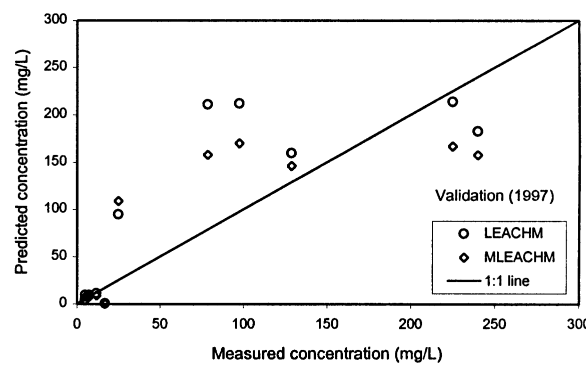
(a) Depth from soil surface = 851 mm



(c) Depth = 1156 mm



(b) Depth = 851 mm



(d) Depth = 546 mm

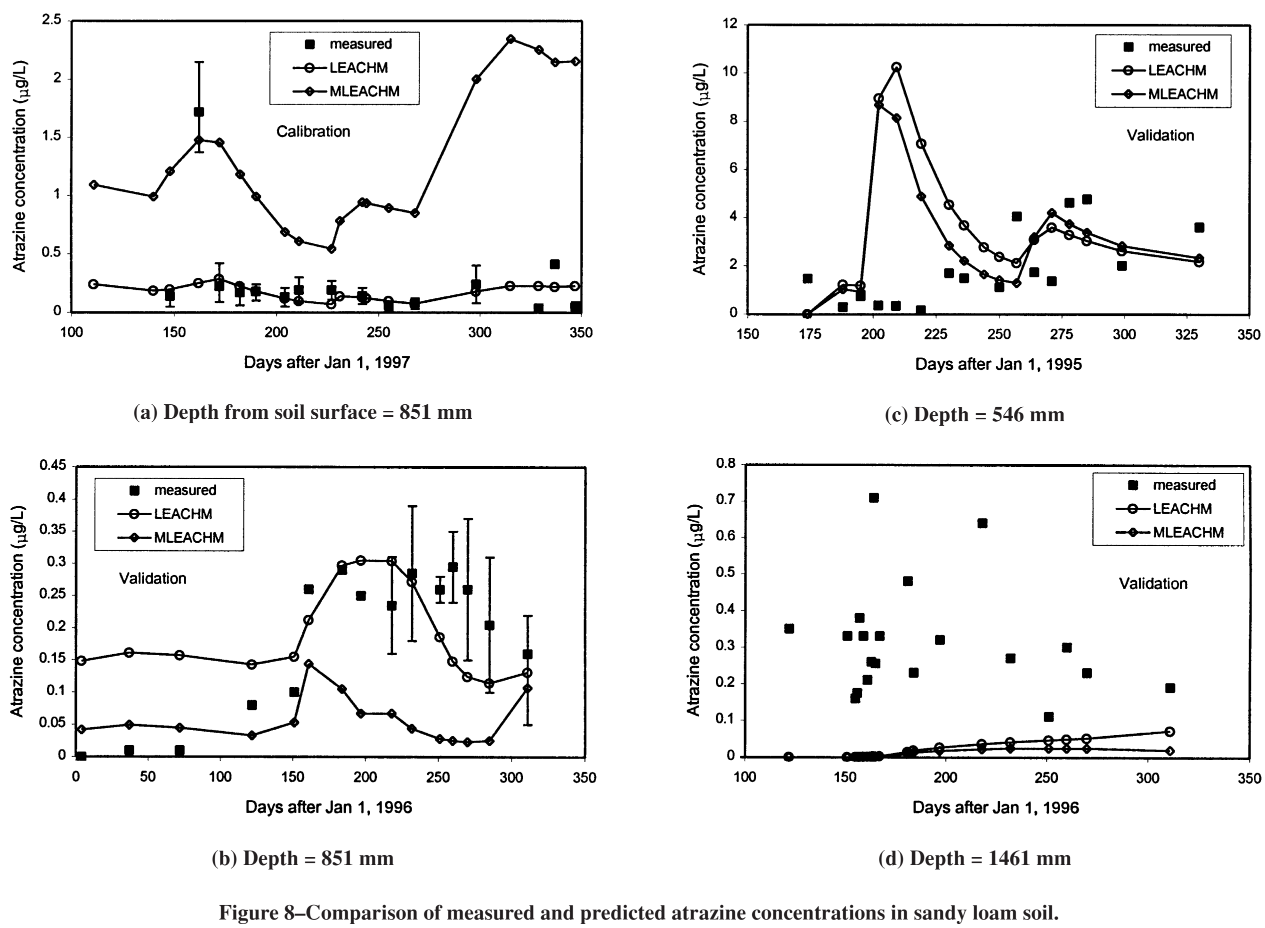
Figure 7–Comparison between measured and predicted nitrate-N concentrations and 1:1 line for silty clay loam soil.

Although, the models predicted a rise in concentration at depth 851 mm, they were unable to predict the rise in concentrations at lower two sampling depths. Figures 6 and 7 present the distribution of measured and simulated NO_3-N concentrations around the 1:1 line.

The models performed similarly in simulating atrazine concentrations in the sandy loam soil (fig. 8). However, both the models overpredicted these concentrations at depth 546 mm and underpredicted at depths 1156 and 1461 mm in 1995 and 1996. Also, MLEACHM overpredicted measured concentrations at depth 851 mm in 1997. In the silty clay loam soil, MLEACHM followed the trend of measured atrazine concentrations in almost all the sampling depths (fig. 9). While LEACHM predictions were good at depth 546 in 1996 and 1997, it predicted almost zero concentrations at all other sampling depths throughout the simulation period. MLEACHM also largely underpredicted at depths 1156 and 1461 mm in 1995. Statistical analyses showed a relatively better performance of MLEACHM in predicting atrazine concentrations in the silty clay loam soil. The distribution of simulated atrazine concentrations against measured concentrations around the 1:1 line are shown in figures 10 and 11.

It can be seen from figure 1 that there were three solute suction samplers at each depth and considerable variations in chemical concentrations at the same depth were observed as indicated by figures 4c, 4d, 5b, 5d, 8b and 9c. The error bars in these figures indicated minimum and maximum values corresponding to the mean values that are plotted as measured data. Error bars could not be plotted for most of the measured data because most of the time water samples were not available from all the samplers at the same depth and at the same time. Although the incorporation of a macropore flow component seems to improve the predictions of LEACHM in silty clay loam soil, especially for atrazine, it was observed that some of the sudden rises in concentrations (which were possibly the results of macropore flow) were not reflected in the simulations. This may be caused by the lack of measured data for macropores (the macropore characteristics were not measured). In this study, a fraction of macropores was estimated at the surface and was gradually reduced with depth. Also, the macropores were assumed as constant throughout the simulation period; whereas, in reality, they change considerably with changes in water content.

The addition of a macropore flow component to LEACHM resulted in only slight improvement in the prediction of NO_3-N concentrations in silty clay loam soil. A possible reason for this may be the high sensitivity of the model to the initial nitrogen and carbon pool in the soil profile, especially litter-N and litter-C, which may reduce the effect of NO_3-N transported through macropores. Some of the peaks in NO_3-N concentrations in sandy loam soil might be the result of preferential flow, but the macropore flow component used in this study may not be adequate for modeling preferential flow in sandy loam soil. Analysis of experimental results also indicated redistribution of chemicals in winter caused by the freezing of the soil which was not accounted for in the models; it may have contributed to the errors in the results. Experimental



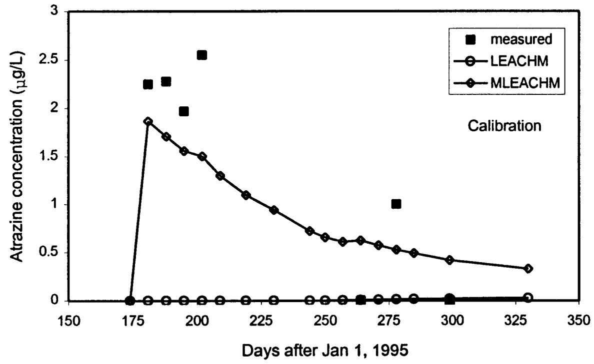
(a) Depth from soil surface = 851 mm

(c) Depth = 546 mm

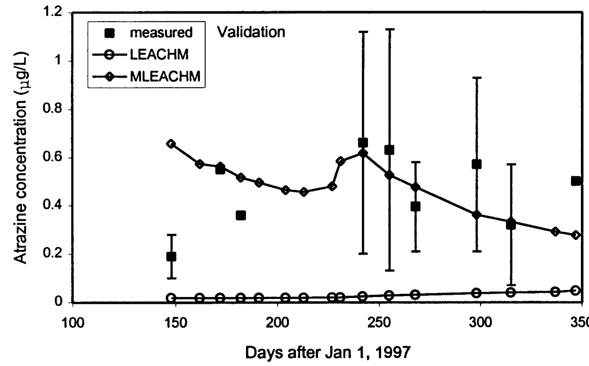
(b) Depth = 851 mm

(d) Depth = 1461 mm

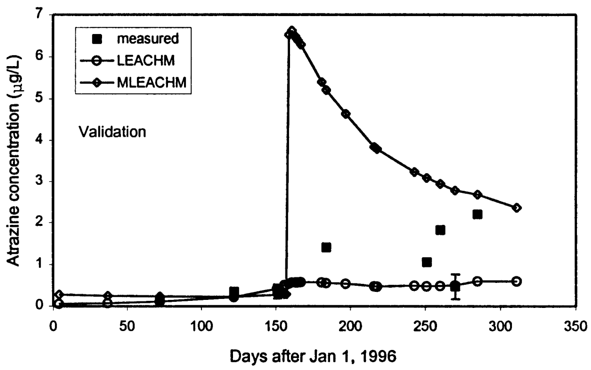
Figure 8–Comparison of measured and predicted atrazine concentrations in sandy loam soil.



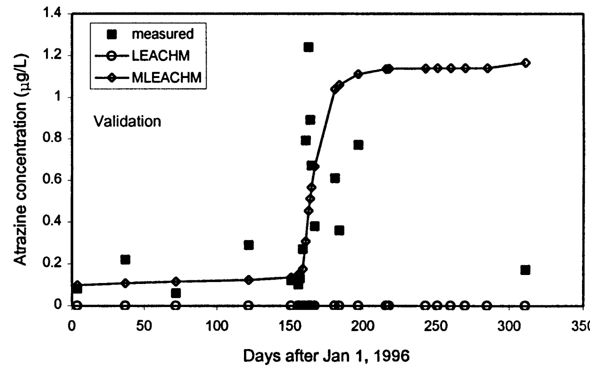
(a) Depth from soil surface = 546 mm



(c) Depth = 851 mm

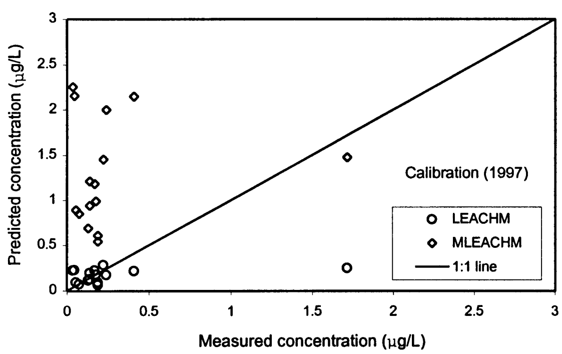


(b) Depth = 546 mm

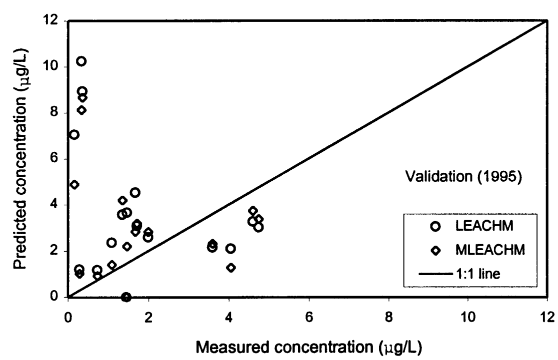


(d) Depth = 1461 mm

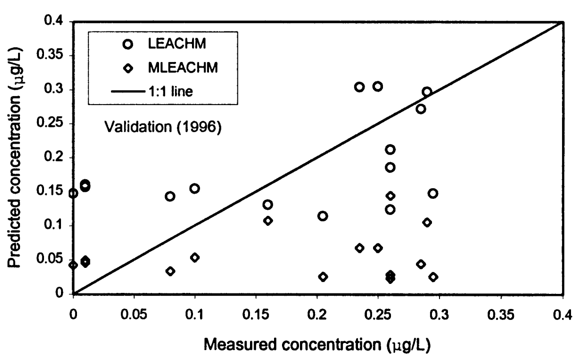
Figure 9–Comparison of measured and predicted atrazine concentrations in silty clay loam soil.



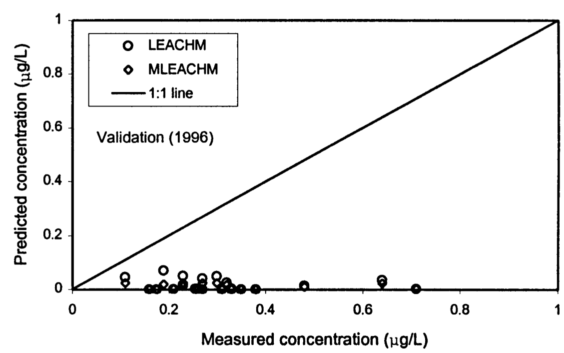
(a) Depth from soil surface = 851 mm



(c) Depth = 546 mm

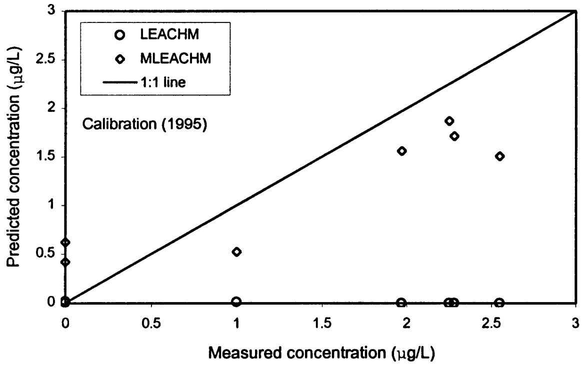


(b) Depth = 851 mm

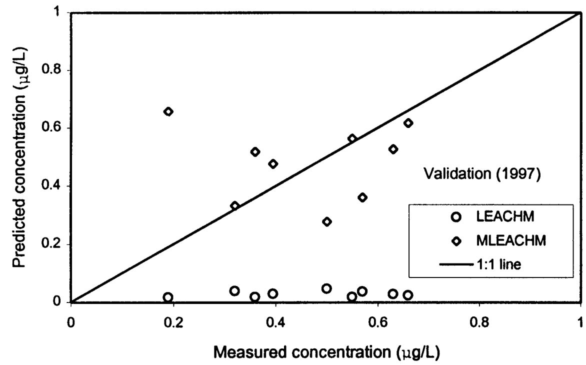


(d) Depth = 1461 mm

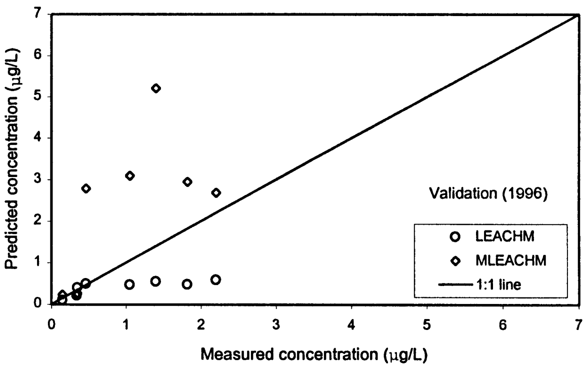
Figure 10–Comparison between measured and predicted atrazine concentrations and 1:l line for sandy loam soil.



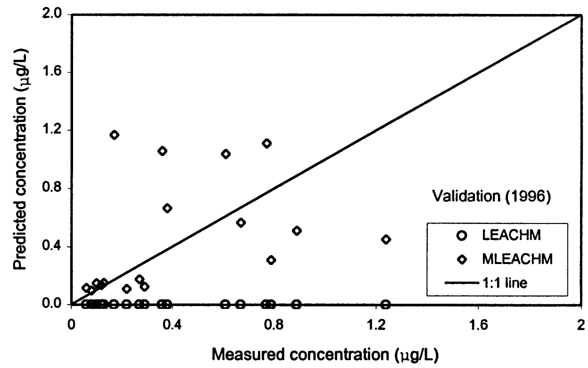
(a) Depth from soil surface = 546 mm



(c) Depth = 851 mm



(b) Depth = 546 mm



(d) Depth = 1461 mm

Figure 11–Comparison between measured and predicted atrazine concentrations and 1:1 line for silty clay loam soil.

NO_3-N concentration data indicated a high rate of mineralization in the late spring. The mineralization rate is dependent upon temperature and moisture content of the soil, among other factors. Since the lysimeters are placed in an underground chamber, the bottom and side walls of the lysimeters are exposed to the atmosphere. As a result, the temperature distribution inside the lysimeters, especially close to the walls, may be slightly different than that in the field and may slightly affect model predictions. Similarly, due to the closed bottom boundary condition of the lysimeters, the moisture content distribution inside the lysimeters may be different than that in the field. Therefore, the mineralization rate and hence the effect of mineralization on NO_3-N concentrations might be different in lysimeters than that in the field. Although, LEACHM takes into account the effect of moisture content and temperature on the mineralization rate constants, it may not represents the physical conditions inside the lysimeters adequately.

CONCLUSIONS

The NO_3-N and atrazine concentrations simulated by LEACHM and modified LEACHM generally followed the trend of measured concentrations. The performance of the models are similar in simulating NO_3-N transport in both the soils and atrazine transport in the sandy loam soil. The original model failed to follow the trend of atrazine concentrations at any of the depths in the silty clay loam soil throughout the simulation period, but the modified model performed better in predicting the trends of atrazine concentrations. This shows the potential for improving the prediction capability of LEACHM in the silty clay loam soil by incorporating a macropore flow component. One possible reason for not observing any significant effect of macropore flow on NO_3-N transport is the high sensitivity of the model to initial carbon and nitrogen pools in the soil profile. There were some discrepancies between measured and predicted values of atrazine concentrations in the silty clay loam soil which might be due to lack of measured macroporosity data.

REFERENCES

ALPKEM Corporation. 1986. RFA Methodology No. A303-S170. Nitrate+Nitrite Nitrogen. Clackamas, Oreg.

Addiscott, T. M., and R. J Wagenet. 1985. Concepts of solute leaching in soils: A review of modeling approaches. *J. Soil Sci.* 36(3): 411-424.

Bogardi, I., and A. Bardossy. 1984. Stochastic forecasting of N budget in the unsaturated zone. In *Proc. of Intn'l. Symp. of Recent Investigations in the Zone of Aeration*, 743-756, Technical University of Munich, Munich, Germany.

Bonazountas, M., and J. M. Wagner. 1984. *SESOIL: A Seasonal Soil Compartment Model*. Cambridge, Mass.: Arthur D. Little.

Brooks, R. H., and A. T. Corey. 1964. Hydraulic properties of porous media, 1-15. Hydrology Paper 3. Fort Collins, Colo.: Colorado State University.

Campbell, G. 1974. A simple method for determining unsaturated conductivity from moisture retention data. *Soil Sci.* 117(6): 311-314.

Canter, L. W. 1996. Nitrates in Groundwater. Boca Raton, Fla.: Lewis Publishers.

Carsel, R. F., C. N. Smith, L. A. Mulkey, J. D. Dean, and P. Jowise. 1984. *Users Manual for the Pesticide Root Zone Model (PRZM)*: Release 1. EPA-600/3-84-109. Athens, Ga.: U.S. Environmental Protection Agency.

Celia, M. A., E. T. Bouloutas, and R. L. Zarba. 1990. A general mass-conservative numerical solution for the unsaturated flow equation. *Water Resour. Res.* 26(7): 1483-1496.

Clemente, R. S., S. O. Prasher, and S. F. Barrington. 1993. PESTFADE, A new pesticide fate and transport model: Model development and verification. *Transactions of the ASAE* 36(2): 357-367.

Crank, J., and P. Nicolson. 1947. A practical method for numerical evaluation of solutions of partial differential equations of heat conduction type. *Proc. Cambridge Phil. Soc.* 43(Part 1): 50-67.

de Willigen, P., L. Bergstrom, and R. G. Gerritse. 1990. Leaching models of the unsaturated zone: Their potential use for management and planning. In *Proc. of the Intl. Symp. on Water Quality Modeling of Agricultural Non-point Sources*. Part 1, ed. D. G. DeCoursey, 105-128. Washington, D.C.: USDA-ARS.

Donnigan Jr., A. S. 1983. Model predictions vs. field observations: The model validation/testing process. In *Fate of Chemicals in the Environment*, eds. R. L. Swann, and A. Eschenroder, 151-171. Am. Chem. Soc. Symp. Ser. 225. Washington, D.C.: ACS.

Enfield, C. G., R. F. Carsel, S. Z. Cohen, T. Phan, and D. M. Walters. 1982. Approximating pollutant transport to ground water. *Ground Water* 20(6): 711-722.

Hutson, J. L., and A. Cass. 1987. A retentivity function for use in soil-water simulation models. *J. Soil Sci.* 38(1): 105-113.

Hutson, J. L., and R. J. Wagenet. 1992. Ver. 3. *LEACHM. Leaching Estimation and Chemistry Model: A process based model of water and solute movement, transformations, plant uptake and chemical reactions in unsaturated zone*. Ithaca, N.Y.: Dept. of Agronomy, Cornell University.

Inskeep, W. P., J. M. Wraith, J. P. Wilson, R. D. Snyder, R. E. Macur, and H. M.Gaber. 1996. Input parameter and model resolution effects on predictions of solute transport. *J. Environ. Qual.* 25(3): 453-462.

Jabro, J. D., J. D. Toth, Z. Dou, R. H. Fox, and D. D. Friton. 1995. Evaluation of nitrogen version of LEACHM for predicting nitrate leaching. *Soil Sci.* 160(3): 209-217.

Kalita, P. K., P. L. Barnes, G. J. Kluitenberg, and A. P. Schwab. 1996. Monitoring pesticide leaching into soil and groundwater through monolith weighing lysimeters. In *Proc. of the Intl. Agricultural Engineering Conf.*, 449-462, Vol. I. Pune, India. Pune, India: Bhumata Prakashan.

Kalita, P. K., G. J. Kluitenberg, P. L. Barnes, A. P. Schwab, J. K. Koelliker, J. M. Steichen, R. Black, M. J. Borah, and D. L. Oard. 1998. A monolith weighing lysimeter for characterizing the fate and transport of agricultural chemicals in soils. *Applied Engineering in Agriculture* 14(5): 485-491.

Kanwar, R. S., H. P. Johnson, and J. L. Baker. 1983. Comparison of simulated and measured nitrate losses in tile effluent. *Transactions of the ASAE* 26(5): 1451-1457.

Khakural, B. R., P. C. Robert, W. C. Koskinen, B. A. Sorenson, D. D. Buhler, and D. L. Wyse. 1995. Test of the LEACHP model for predicting atrazine movement in three Minnesota soils. *J. Environ. Qual.* 24(4): 644-655.

Knisel, W. G., ed. 1980. CREAMS: A field-scale model for chemicals, runoff, and erosion from agricultural management systems. Research Report No. 26. Washington. D.C.: USDA.

Leonard, R. A., W. G. Knisel, and D. A. Still. 1987. GLEAMS: Groundwater Loading Effects of Agricultural Management Systems. *Transactions of the ASAE* 30(5): 1403-1418.

Loague, K., and R. E. Green. 1991. Statistical and graphical methods for evaluating solute transport models: Overview and application. *J. Cont. Hydr.* 7(1-2): 51-73.

Molina, J. A. E., and K. Richards. 1984. Simulation model of the nitrogen and carbon cycle in the soil-water-plant system, NCSWAP; Guide for the preparation of input data files and execution of NCSWAP. Soil Series 116. St. Paul, Minn.: Dept. of Soil Sci., University of Minnesota.

Ohmicron Environmental Diagnostics. Atrazine RaPID assay. Newtown, Pa.

Singh, P., and R. S. Kanwar. 1995a. Simulating NO_3-N transport to subsurface drain flows as affected by tillage under continuous corn using modified RZWQM. *Transactions of the ASAE* 38(2): 499-506.

Singh, P., and R. S. Kanwar. 1995b. Modification of RZWQM for simulating subsurface drainage by adding a tile flow component. *Transactions of the ASAE* 38(2): 489-498.

Smith, W. N., S. O. Prasher, and S. F. Barrington. 1991. Evaluation of PRZM and LEACHMP on intact soil columns. *Transactions of the ASAE* 34(6): 2413-2420.

Steichen, J., J. K. Koelliker, D. Grosh, A. Herman, R. Yearout, and V. Robbins. 1988. Contamination of farmstead wells by pesticides, volatile organics, and inorganic chemicals in Kansas. *Groundwater Monitoring Rev.* 8(3): 153-160.

Thooko, L. W., R. P. Rudra, W. T. Dickinson, N. K. Patni, and G. J. Wall. 1994. Modeling pesticide transport in subsurface drained soils. *Transactions of the ASAE* 37(4): 1175-1181.

USDA-ARS. 1992. Root zone water quality model, Version 1.0. Technical documentation. GPSR Tech. Rept. No. 2. Fort Collins, Colo.: USDA-ARS-GPSR.

USDA-ARS. 1996. Root zone water quality model, Version 3.5. User's manual. GPSR Tech. Rept. No. 6. Fort Collins, Colo.: USDA-ARS-GPSR.

Verma, A., R.S. Kanwar, and U.S. Tim. 1995. Modification of DRAINAGE model by using the nitrogen component from the GLEAMS model. *Transactions of the ASAE* 38(3): 717-724.